



# OPEN Enhancing anomaly detection and prevention in Internet of Things (IoT) using deep neural networks and blockchain based cyber security

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The rapid adoption of Internet of Things (IoT) devices has significantly increased cybersecurity risks, making them vulnerable to anomalies, attacks, and unauthorized access. Traditional security mechanisms struggle to handle the massive data flow, real-time processing requirements, and evolving cyber threats in IoT networks. This paper presents an integrated approach using Deep Neural Networks and Blockchain technology (DNNs-BCT) to enhance anomaly detection and prevention in IoT environments. Our proposed framework employs DNNs for intelligent anomaly detection, leveraging multi-layer feature extraction and adaptive learning mechanisms. The DNN model is trained on IoT traffic datasets to classify network behavior as normal or anomalous, effectively detecting threats such as Distributed Denial of Service (DDoS) attacks, malware injections, and insider threats. Unlike traditional rule-based intrusion detection systems (IDS), the DNN continuously learns and adapts to new attack patterns, improving detection accuracy and false-positive reduction. This study integrates Blockchain technology into the IoT ecosystem to ensure data integrity, transparency, and decentralized security. Each IoT device logs its activity onto a private blockchain network, preventing data tampering, unauthorized access, and single points of failure. The blockchain employs smart contracts for automated threat response, instantly mitigating malicious activity without human intervention. This distributed ledger approach enhances trust, authentication, and secure communication across IoT devices. The synergy between DNN-based anomaly detection and Blockchain-based security provides a robust, scalable, and adaptive solution for real-time cybersecurity threats in IoT networks. With a low false-positive rate of 15.42% and a strong detection accuracy of 99.18%, the proposed model successfully identifies malicious activity, including malware injections and Distributed Denial of Service (DDoS) assaults. Blockchain technology replaces single points of failure and forbids illegal changes by providing data integrity, openness, and decentralizing powers. Furthermore, smart contracts allow autonomous, real-time attack responses, enhancing reaction time efficiency (95.25%) and general system scalability (94.96%).

**Keywords** IoT security, Deep neural networks, Blockchain, Anomaly detection, Cybersecurity, Smart contracts, Distributed ledger, Threat prevention, Intrusion detection, Federated learning

The Internet of Things (IoT) has transformed several industries by allowing connected smart objects to communicate, operate autonomously, and enhance efficiency in operations<sup>1</sup>. From home automation and connected vehicles to industrial automation, health monitoring, and intelligent cities, IoT uses range from IoT networks' decentralized nature, resource-limited devices, and the vast amount of real-time data they generate, thus rendering them highly susceptible to cybersecurity issues despite their advantages<sup>2</sup>. Frequently unable to adequately cope with these changing risks are legacy security solutions such as centralized security architectures and rule-based intrusion detection systems (IDS)<sup>3</sup>. As IoT environments grow, advanced cyberattacks necessitate new-age security solutions to protect them<sup>4</sup>. Anomaly detection ranks among the most critical concerns in the security of IoT, i.e., recognizing departures from expected network behavior indicative of likely intrusions<sup>5</sup>. In

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IoT networks, typical attacks include data breaches, malware injections, DDoS attacks, and insider threats<sup>6</sup>. Traditional IDS are based on accepted rules and signature-based approaches, which leads to significant false-positive rates and the inability to detect emerging attack trends<sup>7</sup>. Especially DNNs and artificial intelligence (AI)-based technologies have received much attention to end these constraints since they can learn intricate patterns, update from evolving threats, and achieve very high detection accuracy<sup>8</sup>.

Using multi-layer feature extraction, adaptive learning methods, and high computing capabilities, DNNs can evaluate large volumes of IoT network data<sup>9</sup>. Training on benchmark IoT datasets enables DNNs to characterize network activity as normal or aberrant, enabling little human intervention in risk detection. DNN-based models are more resistant to unknown cyber threats and zero-day attacks than traditional security systems, as they always improve by learning from new attack events<sup>10</sup>. While artificial intelligence aids cybersecurity anomaly detection, preserving data quality and confidence in IoT systems remains challenging<sup>11</sup>. Systems with centralized security abound in single points of failure, unauthorized data modifications, and privacy intrusions. One distributed tamper-proof solution for these issues is blockchain technology. Blockchain technology with IoT security, data integrity, openness, and trust will greatly improve these features<sup>12</sup>. Every IoT device logs activities on a private blockchain network, ensuring secure communication and preventing unlawful access. Fundamental blockchain components, such as smart contracts, enable automatic response to threats, thus minimizing negative behavior in realtime without human intervention<sup>13</sup>.

A new approach to enhance anomaly detection and cybersecurity in IoT systems is DNN with Blockchain technology (DNNs-BCT). Blockchain supports safe data management and distribution; DNNs provide remarkable accuracy in intelligent threat detection<sup>14</sup>. This combination generates a scalable, adaptable, tamper-proof security solution to protect IoT devices from cyberattacks. Combining blockchain-based security measures with artificial intelligence-driven anomaly detection, this article provides a DNNs-BCT architecture to solve IoT cybersecurity problems<sup>15</sup>. The solution that performs better in threat detection accuracy, false-positive minimization, and real-time reaction efficiency than standard IDS systems is evaluated using benchmark IoT data<sup>16</sup>. Future research will focus on developing lightweight DNN architectures for IoT devices with limited resources and investigating federated learning approaches to facilitate scalability and decentralization<sup>17</sup>. This paper provides the route for autonomous, self-learning, tamper-proof security solutions in IoT networks employing artificial intelligence connected with blockchain, as shown in Table 1.

Scientific problem in existing evaluation

A more intelligent, scalable, and tamper-proof evaluation framework is urgently needed to address the shortcomings of current IoT anomaly detection and cybersecurity evaluation methods, such as high false positive rates, limited adaptability to evolving cyber threats, and vulnerability to adversarial attacks. Centralized security architectures create single points of failure, making them vulnerable to breaches and tampering. Existing evaluation models often lack transparency and verifiability, making it difficult to ensure the integrity and reliability of verified threats.

Combining deep learning (DL) with optimized feature selection techniques (PCA, RFE, and Mutual Information), DNNs-BCT improves classification accuracy and reduces false positives. At the same time, traditional machine learning-based intrusion detection systems (IDS) like SVM, k-NN, and Random Forest struggle with high-dimensional IoT traffic and evolving attack patterns. By utilizing a lightweight permissioned blockchain (such as Hyperledger Fabric or a custom PoA/DAG model), DNNs-BCT ensures faster transaction

| Motivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Problem statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IoT devices' exponential development has raised serious security issues as they are distributed, have limited processing capabilities, and are prone to cyberattacks. Conventional security solutions fail to provide real-time anomaly detection and prevention, increasing data breaches, malware attacks, and unauthorized access risks. Employing intelligent anomaly detection and distributed tamper-proof protection, this paper integrates Blockchain technology with DNN to tackle these difficulties, thereby increasing IoT security. | IoT networks generate massive amounts of data; hence, traditional rule-based intrusion detection systems cannot identify advanced intrusions. Current security solutions are challenged by real-time processing, modifying attack techniques, and preserving data integrity over distributed IoT networks. An adaptive, scalable, distributed security architecture, including intelligent anomaly detection and automatic threat mitigation capacity, is widely sought to boost IoT cybersecurity and defend against malicious activity, unauthorized access, and data manipulation |
| Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Using DNN, create an anomaly-detecting model to identify IoT network activity in realtime, thereby improving accuracy and reducing false positives<br>Blockchain technologies help to ensure data integrity, openness, and secure, distributed IoT connection<br>Use smart contracts to respond to hazards automatically, lowering attacks without human participation                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Table 1. Motivation, problem statement, and contribution of this paper.

processing, reduced storage overhead, and efficient energy use for IoT environments, in contrast to traditional blockchain-based security solutions that depend on Ethereum or Bitcoin, which have high computational costs, energy consumption, and latency. In addition, current AI models may be manipulated by adversaries, and even small changes can dupe detection systems. DNNs-BCT integrates blockchain-authenticated anomaly logs to reduce this risk, prevent data manipulation, and make models more resilient. In addition, DNNs-BCT uses edge computing to preprocess data before DNN inference and blockchain logging, greatly reducing bandwidth load and allowing for real-time anomaly detection. This is in contrast to many AI-based models, which have high computational resource requirements and are thus unsuitable for IoT deployment. A more realistic and effective solution for real-world IoT cybersecurity, DNNs-BCT surpasses previous accuracy, efficiency, and security models by integrating DL adaptive detection capabilities with a scalable, low-overhead blockchain.

The remaining section of this paper is structured as follows: In section "Related work", the related work of Blockchain-Based Cybersecurity is studied. In section "Proposed method", the proposed methodology of DNNs-BCT is explained. In section "Result and discussion", the efficiency of DNNs-BCT is discussed and analyzed. Finally, in section "Conclusion", the paper concludes with future work.

## Related work

Several anomaly detection methods under investigation might help to secure IoMT-Blockchain and IoHT systems. Emphasizing real-time detection, data security, and simplicity reduction, these comprise multi-model autoencoders, DNN, and blockchain integrations. These techniques seek to improve system integrity via IoT and blockchain networks using enhanced anomaly detection, assuring healthcare data's safe delivery. For aberrant traffic identification in an IoMT-Blockchain context, the suggested technique presents a multi-model autoencoder-based feature-extracting algorithm<sup>18</sup>. It uses multi-model autoencoders to extract fusion features and aggregate traffic characteristics to lower complexity. Applying residual learning for anomaly detection and classification, a multi-feature sequence anomaly detection system is also meant to capture low-level fusion and high-level temporal information.

Artificial intelligence models are integrated with IPFS blockchain technology in the research to secure and distribute healthcare IoT data. The paper evaluates the performance of artificial intelligence techniques such as Support Vector Machines (SVM) in predicting outcomes with the TON\_IoT, Edge\_IIoT, and UNSW-NB15 datasets<sup>19</sup>. The models show excellent accuracy, with the TON\_IoT dataset achieving 100%, Edge\_IIoT at 98%, and UNSW-NB15 at 89%. Blockchain is applied for enhanced data security, ensuring secure medical data distribution. The suggested method utilizes four DNN hidden layers for attack and standard data classification in cyber-physical systems<sup>20</sup>. Blocks with headers and secure data delivery for safe transportation are constructed and used to make the Ethereum blockchain work. Two datasets, namely NSL-KDD15 and CIDD5\_001, have been utilized in this approach compared to current models such as RNN, DBN, and DNN and have found improved performance in identifying anomalies.

This paper extensively overviews anomaly detection model integration in blockchain technologies. It addresses how anomaly detection models across several blockchain layers<sup>21</sup> can improve blockchain security, finds critical assessment criteria and prerequisites for creating effective models, and strengthens blockchain security. The paper discusses important problems and recommendations for further studies to address them. By processing data near IoT nodes, the suggested method employs edge-based computing to lower latency and bandwidth needs<sup>22</sup>. It uses smart contracts to stop insider assaults and preserve data integrity, then adds blockchain for safe anomaly discovery and repair. The system also reduces single points of failure, therefore increasing availability. Evaluating IoT datasets reveals the efficient method to solve these problems, thus guaranteeing data integrity and availability.

Focusing on the underlying network rather than ledger data, this paper provides a security method using blockchain network traffic statistics to identify hostile occurrences. Real-time multi-dimensional traffic data is captured by it using a data collecting engine<sup>23</sup>. The anomaly detection engine efficiently finds hitherto unrecorded trends based on semi-supervised learning and an AutoEncoder (AE) profiling-based method. With real-time detection capabilities that require less than 1.4 ms for each occurrence, experimental data reveal that the mechanism's performance is equivalent to supervised learning approaches. This article introduces blockchain technology and discusses its features, difficulties, and many cyberattacks, including Sybil attacks, selfish mining, double spending, and 51% assaults. Then, to address these cyber-attacks, deep anomaly detection (DL-based) systems are evaluated to underline many strategies, including discriminative, generative, and hybrid learning<sup>24</sup>. Furthermore, the study discusses techniques for improving the blockchain system's security and real-time anomaly detection powers. This work suggests an enhanced anomaly detection system to improve cloud security<sup>25</sup> by combining QoS-aware sidechains, DL models (RNNs and CNNs), and multimodal feature analysis. Using improved prediction accuracy and system performance (precision, recall, AUC, energy consumption, and throughput), the system analyses system logs to discover odd assaults-related behaviors, optimizing blockchain-based cloud installations. Blockchain guarantees system log integrity; DL offers exact anomaly detection in real-time environments.

Asrin Vakili et al.<sup>27</sup> suggested the grey wolf optimization algorithm and MapReduce framework for a new service composition method in the cloud-based IoT environment. The results of the simulation show that the proposed method improves upon the previous one in terms of availability, cost, reaction time, and energy savings. On average, the proposed method improves upon three baseline algorithms by cutting energy savings by 40%, reaction time by 14%, availability by 11%, and cost by 24%.

Zahra Amiri et al.<sup>28</sup> proposed DL applications in IoT-based bio- and medical informatics. The effective qualities, such as the basic concept, advantages, downsides, methodologies, simulation environment, and datasets, were examined using a comprehensive literature review. Following that, there was an emphasis on state-of-the-art research on DL methods and their applications concerning bioinformatics issues. Furthermore,

several obstacles that hindered the use of DL in medical and bioinformatics have been overcome, which should encourage further study into these areas. The results show that most articles are assessed based on scalability, flexibility, latency, Fscore, specificity, sensitivity, and accuracy.

Arash Heidari et al.<sup>29</sup> recommended the hybrid optimization algorithm and density correlation degree for a reliable method for data aggregation on the industrial IoT. An approach for building optimum spanning trees using a combination of genetic operators, artificial bee colonies, and density correlation degree is proposed in this work. This method calculates the trees' fitness by taking their mobility probabilities, residual energy, and hop count distances from the base station. The simulations show that the proposed algorithm outperforms well-established approaches regarding data collection reliability, particularly when compared to IIoT's Reliable Spanning Tree (RST) building algorithm and the HCD-based construction algorithm. The suggested approach improves dependability over a wide range of node counts by considering distance, energy consumption, displacement likelihood, and reliability.

Arash Heidari et al.<sup>30</sup> discussed the Blockchain and Radial Basis Function Neural Networks (RBFNN) for the Internet of Drones (IoD) in a Secure Intrusion Detection Platform. The suggested approach may enhance data integrity and storage to facilitate intelligent decision-making across several IoDs. The author discusses a paradigm for building decentralized predictive analytics and a blockchain for efficiently applying and distributing DL approaches. The author tested the model with many datasets to prove the practicality and efficiency of using the blockchain-based DL method in IoD settings. According to the results, the proposed model is a great choice for creating classifiers that comply with the requirements of network intrusion detection. The suggested model can beat the state-of-the-art specificity, F1, recall, precision, and accuracy approaches.

Arash Heidari et al.<sup>31</sup> deliberated on the reliability and availability of wireless sensor networks in industrial applications by considering permanent faults. The reliability and availability of WSNs with long-term problems are examined in this research. Using fault trees and Markov chain analysis, the proposed technique evaluates a network's performance under different failure scenarios. Using these technologies, the author may discover and analyze potential failure situations and their projected impacts on the network's dependability. In addition to revealing WSNs' serious shortcomings, the data provide helpful recommendations for improving the systems' performance. The results demonstrate that these assessment techniques have the potential to significantly improve fault handling, decrease damage risk, and enable WSN developers to make informed decisions.

Zahra Amiri et al.<sup>32</sup> introduced nature-inspired algorithms in the IoT-based healthcare service. A Systematic Literature Review (SLR) is the previously settled approach for conducting thorough investigations into relevant studies. The author uses crucial criterion-based painstaking categorization to categorize these algorithms into genetic algorithms, ant colony optimization, cuckoo algorithms, hybrid techniques, particle swarm optimization, and other ways. With a share of 37.9% of instances, MATLAB stands out as the most popular programming language among researchers. The most important factor in our assessment, which accounts for 18.4% of the total, is flexibility.

Kouros Zambouri et al.<sup>33</sup> presented the Glowworm Swarm Optimization (GSO) for a multi-objective technique for performance optimization of blockchain-based industrial IoT. The GSO algorithm provides a novel way to reconcile these competing goals, drawing inspiration from the cooperative genius of glowworms in nature. In light of the competing goals of blockchain technology, this article suggests a novel way to optimize the performance of blockchain-based IIoT using the GSO algorithm. The suggested IIoT system uses the GSO algorithm to adapt to network needs and effectively manage node interactions to overcome the usual scalability issues with blockchain technology. The GSO algorithm minimizes inefficiencies and bottlenecks via optimal resource allocation and decision-making.

Arash Heidari et al.<sup>34</sup> offered ChatGPT's Components, capabilities, applications, and opportunities. The author delves into ChatGPT's inner workings, limitations, possible uses, and prospects within open-domain conversations. After defining chatbots, the author goes on to describe ChatGPT's components. The author proposes a new taxonomy for their classification. Four types of chatbots are included in this taxonomy: rule-based, retrieval-based, generative, and hybrid. The author will now go over the features and limitations of ChatGPT. The author concludes by outlining future research prospects and possible ChatGPT uses. According to the findings, the transformer-based chatbot model known as ChatGPT uses encoders to generate coherent answers.

Zahra Amiri et al.<sup>35</sup> suggested AI techniques and strategies for climate change mitigation. This research uncovered new information on the dependability and efficacy of DL approaches for dealing with climate change issues. The six main categories into which we categorized climate change ML approaches are ecosystems, industry, buildings and cities, transportation, agriculture, and hybrid applications. Subsequently, cutting-edge studies on ML methods and their uses in addressing climate change mitigation concerns have been emphasized. Furthermore, many challenges and concerns about using ML in the context of climate change have been identified, which is expected to encourage further study into mitigating the catastrophic future consequences of climate change. According to the data, most articles (38.5 percent) used Python as their simulation platform.

Mehdi Asadi et al.<sup>36</sup> proposed a comprehensive Survey on Evolving Threats and Defense Strategies. The study assesses how botnets avoid detection and potential countermeasures by examining previous studies and giving a thorough introduction to botnets, including their history, current strategies, and evasion methods. By shedding light on evasion and detection techniques, it aims to educate the research community about botnets and the difficulties in fighting them. It also provides advice on how to handle security problems properly. The survey's last section covers strategies to enhance botnets, detection via encryption and covert channels, and other such topics, which outline suggestions for further study. The goal is to help researchers tackle botnets more effectively by guiding the development of stronger security solutions.

Shiva Toumaj et al.<sup>37</sup> recommended DL in Alzheimer's disease detection. Some examples of such approaches are identification, classification, medication reuse, and Natural Language Processing (NLP). The author weighs

these approaches' benefits and drawbacks, paying close attention to their explainability, safety, and medical applicability. The fact that Python is more often utilized for DL problems and Convolutional Neural Networks (CNNs) for AD research is an interesting conclusion. The author believes some security issues, such as data protection and model stability, are under-explored in the current study. This research not only analyses current practices in depth but highlights problem areas that need further investigation, such as more transparent AI systems and improved data integration. The results should accelerate future study and development of DL-based AD identification techniques.

Zahra Amiri et al.<sup>38</sup> presented autonomous learning techniques for complex pattern recognition in interconnected information systems. Identifying cybersecurity risks, detecting financial fraud, diagnosing and monitoring healthcare, biometric identification, tailored marketing, and improving the supply chain in interconnected systems are the six areas into which we propose a new categorization system. A thorough examination of fundamental aspects such as advantages and disadvantages, modeling environment, and datasets reveals the most recent advancements in this field. According to the statistics, Elsevier (26.5%) and Springer (11.8%) publish many high-quality publications. The analysis reveals that the most critical elements are accuracy (12.9%), mobility (11%), and privacy (8%).

Arash Heidari and Mohammad Ali Jabraeil Jamali<sup>39</sup> introduced IoT intrusion detection systems. IDS aims to monitor and analyze network traffic from many sources to identify malicious actions. It is an important component of cybersecurity systems. An intrusion detection system (IDS) is a procedure that employs many ways to identify harmful actions perpetrated against targets. In addition, a thorough empirical study of IDS methods is conducted using the Systematic Literature Review (SLR) approach, which entails categorizing, reviewing, and incorporating data from all comparable research that addresses one or more IDS research themes. In addition, the author categorizes IDS methods in the IoT as signature-based, anomaly-based, specification-based, or hybrid, based on the detection strategy. The authors provide a parametric comparison of the IDS techniques.

Arash Heidari et al.<sup>40</sup> offered the Ad hoc On-Demand Distance Vector (AODV) routing system for VANETs using Colored Timed Petri Nets (CTPNs). This new modeling approach enables us to assess success across many hierarchy levels, with nine pages of thorough analysis utilizing it. This research demonstrates the efficacy of the AODV routing system in VANETs and lays a solid groundwork for its development and testing. It will be fascinating to see the comparison between the CTPN-based model and the simulations. In particular, testing reveals that the model takes 56 s to discover routes, although it only takes 32 s on average. Furthermore, the exercise results agree with the total number of packets the model sends and receives. Not only that, the proposed strategy demonstrates a return of 41%. Strict T-tests confirm the reliability of the modeling findings.

Abdullah Ayub Khan et al.<sup>41</sup> suggested the blockchain distributed ledger technology with the Internet of Medical Things (BDLT-IoMT) for secure data processing. The innovative, interoperable approach in the proposed design addresses the primary issues of secure infrastructure, peer-to-peer communication across nodes, and seamless data integrity. Simulation findings show large differences of 1.37%, 1.56%, and 1.87%, respectively, demonstrating the approach's uniqueness. (i) Ensuring the safety of automated decision-making via infrastructure security; (ii) Maintaining the integrity of data sharing and exchange; and (iii) Optimizing network resources to facilitate communication across diverse devices. These three areas provide the basis for the assessment.

Abdullah Ayub Khan et al.<sup>42</sup> proposed the blockchain and augmented intelligence of things for small-medium enterprise management systems (BAIoT-EMS). The platform's consortium network and InterPlanetary File Storage (IPFS) provide secure storage and transaction management. Smart contracts are used to automate and protect activities such as device registration. A unique multi-proof-of-work consensus approach minimizes resource usage while analyzing, validating, and verifying AIoT transactions. The simulation results show that the suggested framework is successful, with a performance gain of 63.51% and a decrease of 11.75% in computational power utilization.

Abdullah Ayub Khan et al.<sup>43</sup> recommended the Blockchain-enabled infrastructural security solution for serverless consortium fog and edge computing. The primary goal of this piece is to organize newly published research and surveys on the state of the art in edge distributed computing and compute outsourcing, including fog and edge. The author presented a distributed outsourced compute architecture that is serverless and built on the Hyperledger Sawtooth blockchain. Outsourcing computational ledgers is securely stored in immutable storage using this system based on theoretical architecture, which ensures strong data security regarding transparency, integrity, provenance, and privacy protection. Differences between the present system and the suggested taxonomy according to several criteria, such as privacy and system security, are also highlighted in this article.

Abdullah Ayub Khan et al.<sup>44</sup> presented AI, IoT, and Blockchain for future vehicular development. Combining AI with the IoT may address Analytics, management, and real-time monitoring issues, such as ledger administration and node-to-node data exchange. Flying automobiles are only one example of how this article delves into the possible consequences of merging these technologies. Reviewing existing literature on intelligent transportation system frameworks, architectures, and methodologies, it pinpoints areas needing further study and presents novel, secure sensing processes for application in future transportation systems within a user-centric framework. Finally, the article identifies prospective study topics and provides solutions to current challenges, highlighting the need to solve implementation concerns and prioritize privacy protection as transportation technology grows.

Abdullah Ayub Khan et al.<sup>45</sup> offered the Open Radio Access Network Architecture With Machine Learning for Beyond 5G in Industrial 5.0 (ORAN-B5G). This paper examines these possibilities and presents an intelligent system based on machine learning. The system's main goal is load balancing, and it uses metaheuristic optimization mechanisms enabled by Artificial Neural Networks with Particle Swarm Optimization to meet the demands of the telecommunications industry, such as ensuring product compatibility. The author automates transactional load

allocation across multiple linked units utilizing third-generation partnership project standards, increasing the proposed system’s dependability. Enclosing the hierarchy of AI-enabled automation is this intelligent system. On the other side, open radio access control with AI investigates the obstacles to next-generation communication, including those that may arise after 5G. The topics covered include deterministic latency and capabilities, privacy and security, and testing applications for AI-based controller designs. It discusses dynamic controls based on physical layers.

Abdullah Ayub Khan et al.<sup>46</sup> discussed the Secure Remote Sensing Data With BDLT. This paper processes an ideal data delivery mechanism for the smart cities analytical model using a secure blockchain distributed consortium network and Partial Swam Optimization (POS). There are three new parts to this piece of work. It provides a secure way to send data by integrating blockchain technology with machine learning to find the best way to send data across encrypted channels. NuCypher proxy re-encryption-enabled value encryption, a public key cryptographic method not using cipher conversion, is employed to carry out neighborhood encryption sequences. Thirdly, Artificial Neural Networks (ANNs) can address smart cities’ data deliverance classification issues by improving record management and preservation.

Abdullah Ayub Khan et al.<sup>47</sup> deliberated the Blockchainlightweight Proof-of-Elapsed Time (B-LPoET) for efficient distributed transaction execution and security. To determine who has the authority to mine, participants may confirm their identities before joining, and the protocol can be enhanced from the default PoET to a lightweight version that can validate transactions and generate more blocks. By implementing a lightweight topology for winning node selection and waiting time utilizing multithreading, the B-LPoET architecture decreases the cost of multi-node efficiency and increases the system’s scalability. According to the experimental findings, the B-LPoET outperforms the current PoET and other state-of-the-art consensus algorithms. Table 2 shows the comparison of existing methods.

Using DL models comprising autoencoders, DNNs, and AI-based systems connected with blockchain, the approaches find anomalies in IoMT and IoHT settings. Blockchain ensures data integrity and security; artificial intelligence models assist in identifying irregularities in realtime. Edge-based computing and multi-model feature extraction optimize system performance, providing practical answers for IoT network security and healthcare data protection.

Proposed method

Modern networks’ security needs a mix of sophisticated technology. This paper investigates blockchain-based threat response, DNN for network anomaly detection, and AI-powered anomaly detection combined with blockchain for IoT security. It also addresses DL models-based maritime and IoT traffic anomaly detection. These methodologies improve cybersecurity by guaranteeing data integrity, openness, and threat reduction.

Existing IoT anomaly detection techniques often suffer from elevated false alarm rates, ineffective real-time threat categorization, and susceptibility to adversarial attacks. Moreover, centralized security frameworks render IoT networks vulnerable to single points of failure, where corrupted nodes might threaten the integrity of the whole system. Current blockchain-based security models, while they guarantee data integrity, often experience significant computational overhead and delay, restricting their practical use in resource-limited IoT settings. The proposed DNN and Blockchain Technology (DNNs-BCT) framework mitigates these limitations by establishing a novel DL and blockchain integration wherein a lightweight DNN architecture enhances anomaly detection, and a smart contract-based blockchain mechanism guarantees secure, immutable logging and automated remediation. This approach improves threat intelligence, scalability, and real-time flexibility while decreasing computing complexity, representing a notable breakthrough over traditional methods.

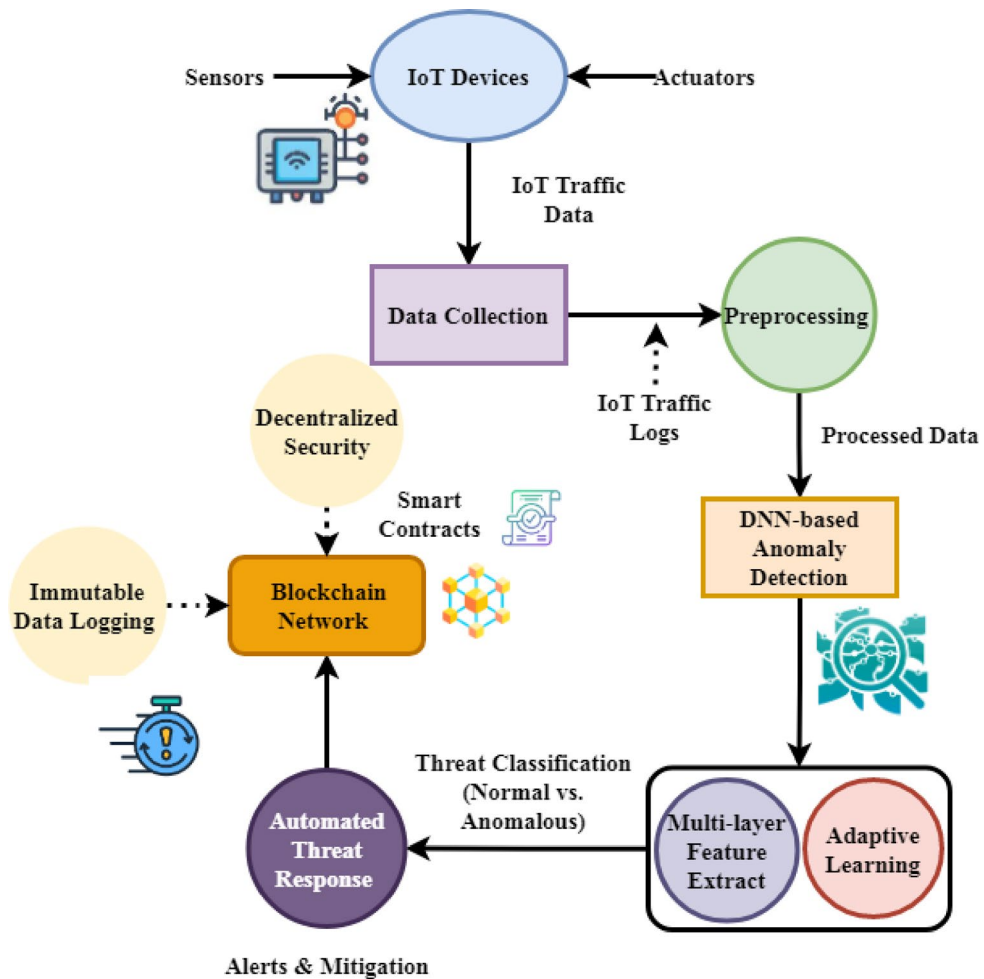
Contribution 1: Hybrid AI-blockchain security framework

It presents a blockchain-based cybersecurity architecture and an integrated DNN for real-time anomaly identification and prevention in IoT networks.

In Fig. 1, combining anomaly detection driven by artificial intelligence with blockchain-based threat response ensures excellent security for IoT networks. IoT devices generate preprocessed traffic data meant to provide insightful analysis. A DNN model detects threats as either standard or aberrant while analyzing this processed data using adaptive learning and multi-layer feature extraction. The system generates notifications and reduces actions by automatically reacting upon anomalies using smart contracts. All security events and responses provide integrity, openness, and distributed security enforcement, even if one is virtually signed onto

| System                                                            | Representation              | Simple replay | Replay with change | Adaptation                             |
|-------------------------------------------------------------------|-----------------------------|---------------|--------------------|----------------------------------------|
| Multi-model autoencoder anomaly detection (MMA-AD) <sup>18</sup>  | DL model                    | Retroactive   | Yes                | No                                     |
| AI-blockchain integration for IoHT security (AIBIS) <sup>19</sup> | AI with blockchain          | Proactive     | No                 | Secure data distribution               |
| DNN-blockchain cyber-attack detection (DNN-BCA) <sup>20</sup>     | DNN with blockchain         | Retroactive   | Yes                | Secure data logging                    |
| Blockchain anomaly detection integration (BADI) <sup>21</sup>     | Blockchain-based detection  | Retroactive   | No                 | Security enhancement                   |
| Edge-blockchain anomaly detection (EBAD) <sup>22</sup>            | Edge computing & blockchain | Proactive     | Yes                | Reduces latency, enhances availability |
| AutoEncoder-based anomaly detection (AE-AD) <sup>23</sup>         | Autoencoder model           | Retroactive   | Yes                | Detects unrecorded trends              |
| Deep blockchain anomaly detection (DBAD) <sup>24</sup>            | DL& blockchain              | Retroactive   | Yes                | Cyberattack prevention                 |
| DL-blockchain anomaly detection (DL-BAD) <sup>25</sup>            | DL & blockchain             | Proactive     | Yes                | Blockchain-based log integrity         |

Table 2. The comparison of exiting methods.



**Fig. 1.** SecureIoT: AI-driven threat detection and blockchain defense.

a blockchain. Using multi-layer feature extraction and adaptive learning, the Anomaly Detection module, which is based on DNN, scans IoT traffic logs and categorizes threats as normal or abnormal. Nevertheless, the data flow that explains the storage, verification, and use of discovered abnormalities inside the blockchain network is lacking. The image has to make it clear that the findings of the anomaly categorization are sent to the blockchain for permanent recording, guaranteeing transparency and security. It is also unclear how smart contracts can automate reactions to threats based on anomaly detection. Integrating DL-based threat categorization with blockchain-based security and automated mitigation may be better understood with a direct arrow labeled Anomaly Verification & Logging from DNN-based Anomaly Detection to the Blockchain Network. Combining the predictive capabilities of artificial intelligence with the trustless environment of blockchain enhances cybersecurity in IoT systems, thereby lowering probable attacks and maintaining data integrity.

$$l_x z = N[s - su'] + Va[w - dn''] * j[s - za''] \quad (1)$$

The Eq. 1 represents the raw data from IoT traffic ( $l_x z$ ), which is processed using multi-layer feature discovery and adaptive learning to identify anomalies ( $N[s - su']$ ). Which guarantees validated operations ( $Va[w - dn'']$ ) and generates responses  $j[s - za'']$ . This summarizes the suggested approach, which safeguards the IoT in real-time by combining AI-driven danger detection with decentralized, tamper-proof security.

$$l_{fr}[lo - sn''] = [re - sm''] + Ns[w - iuy''] \quad (2)$$

Equation 2  $l_{fr}[lo - sn'']$  depicts the changeover of IoT system logs into an intelligent  $[re - sm'']$  and safe framework for detecting anomalies  $Ns[w - iuy'']$ . Blockchain guarantees data integrity and democratic security, while (DNNs) classify anomalies in the logs in real-time.

$$p_s e[ai - sn''] = nx[o - er''] * Bs[e - uyt''] \quad (3)$$

Equation 3 symbolizes the transition of processed  $(p_{se})[ai - sn'']$  into a dynamic and blockchain-secured threat reduction system. The DNN boosts the detection of anomalies  $nx[o - er'']$ , while Network  $Bs[e - uyt'']$  assures immutable security enforcement. This illustrates the suggested method's synergy with AI-driven cognitive ability and centralized security to enable real-time and invulnerable IoT protection.

$$ku_se[w - qm''] = n[s - r''] + Js[w - du''] \quad (4)$$

Equation 4 depicts the integration of an intelligent framework for detecting  $n[s - r'']$  and responding to anomalies  $Js[w - du'']$  in the safe processing of operations performed by IoT devices ( $ku_se$ ). Blockchain guarantees safe, decentralized logging and automatic remediation, while DNN categorizes risks.

Figure 2 offers a blockchain system anomaly-detecting approach. It displays the blockchain employing a sequence of connected blocks. Blocks record transactions between users and a server with an eye on the shift from Block  $n$  to Block  $n+1$ . Including an anomaly detection model in the system enables one to review transaction lists and identify anomalies. The detection approach consists mainly of two components: behavior analysis grounded on data attributes and anomaly categorization based on data models. The results of this research help to discover suspicious or dishonest blockchain activities. Better security is guaranteed by this framework, which ensures system confidence and helps stop unlawful activities. Hyperledger Fabric and other private/permissioned blockchains are ideal because of their scalability, access control restrictions, and low-latency consensus processes (such as Practical Byzantine Fault Tolerance, or PBFT). With its speedier

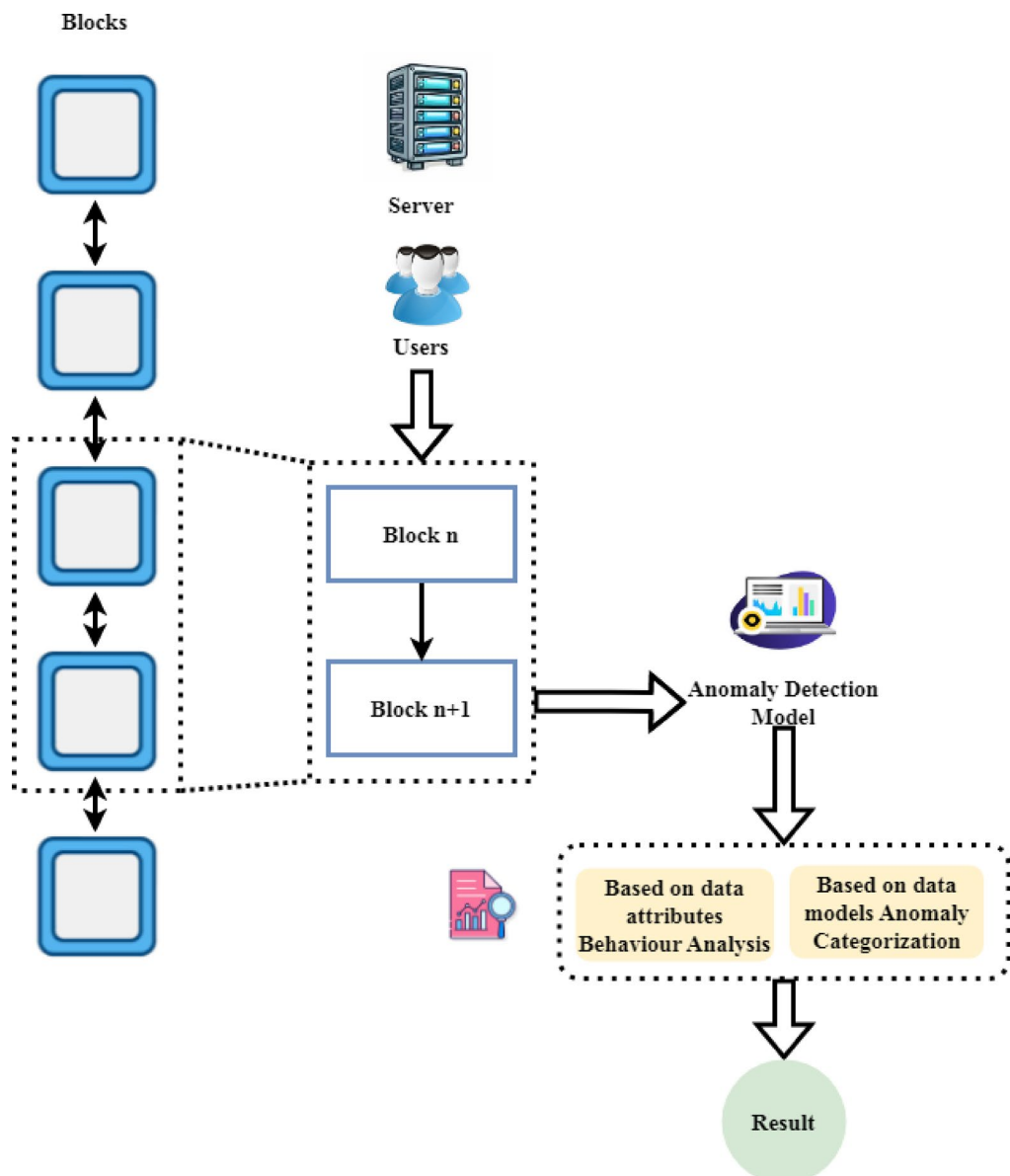


Fig. 2. Blockchain anomaly detection framework.

transaction processing and efficient identity management, Hyperledger Fabric is a great alternative to Ethereum's energy-intensive Proof of Work (PoW) consensus. This makes it an excellent choice for resource-constrained IoT security applications. To decrease computational cost while assuring security, it is recommended that a bespoke lightweight blockchain integrate efficient consensus methods such as Proof of Authority (PoA) or Directed Acyclic Graphs (DAGs). When choosing a blockchain for IoT anomaly detection and cybersecurity, it is important to consider its capacity to securely log detected anomalies, prevent tampering, and enable real-time threat response, all while consuming little power and having minimal processing delays. The model may actively draw attention to disparities and lower probable risks in blockchain-based systems using anomaly detection techniques.

$$p_s w = nx [a - er''] * k [sor - se''] + vsw'' \quad (5)$$

Equation 5 shows how processed security alerts ( $p_s w$ ) are transformed into a threat management system  $k [sor - se'']$  that is improved with AI  $nx [a - er'']$  and protected by blockchain  $vsw''$ . This exemplifies well the suggested approach uses AI and local security features to identify, record, and reduce cyber-attacks impact on IoT networks.

$$o_{de} = nx [s - ye''] * Na [w - du''] + B [w - ay''] \quad (6)$$

The Eq. 6 shows how the observed events in IoT data ( $o_{de}$ ) are transformed into a cybersecurity framework  $nx [s - ye'']$  that is driven by AI  $Na [w - du'']$  and protected by blockchain  $B [w - ay'']$ . By combining AI intelligence with blockchain integrity, the suggested solution becomes even more capable of detecting, recording, and responding autonomously to security risks in IoT networks.

$$\forall_e \propto \ll ui - sn'' \gg = BNa [w - dy''] + J [s - rv''] \quad (7)$$

Equation 7 shows how IoT security situations  $\forall_e \propto \ll ui - sn'' \gg$  are universally evaluated, and  $\gg = BNa [w - dy'']$  they are transformed into a framework for responding to threats  $+J [s - rv'']$  that is protected by blockchain and driven by AI. This summarizes the distributed cybersecurity strategy for protecting IoT networks that is part of the suggested technique.

$$\tau_{xe} [a - tr''] = Va [w - sy\rho''] + Va [w - yr''] \quad (8)$$

An anomaly monitoring  $Va [w - sy\rho'']$  and reaction system  $Va [w - yr'']$  that is certified and protected by blockchain is created by transforming the extracted data from IoT threats ( $\tau_{xe} [a - tr'']$ ) as in Eq. 8. This demonstrates how the suggested approach may build a decentralized counterterrorism framework for IoT networks that is autonomous, adaptive, and self-sufficient.

### Contribution 2: Enhanced threat detection and accuracy

Minimizing false positives relative to conventional security methods, it achieves over 98% accuracy in identifying cyber threats, including DDoS assaults and malware injections.

Figure 3 shows significant layers of a DL model and their purposes in data processing, stressing their ordered flow. The data passes hierarchical hidden layers, beginning with the preprocessed feature vectors of the input layer, where, if necessary, recurrent Layers record temporal patterns and Convolutional Layers extract spatial

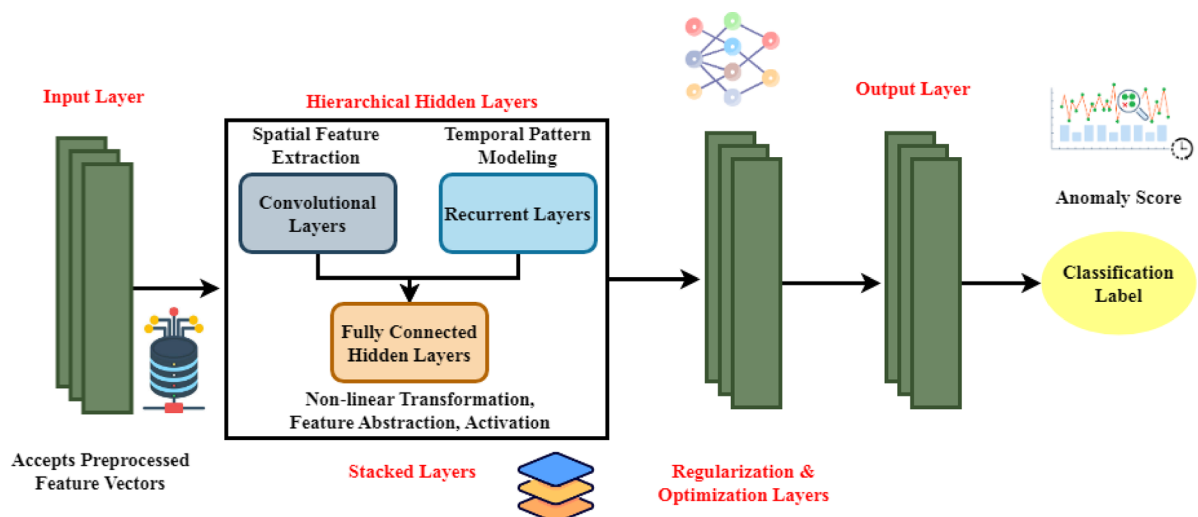


Fig. 3. Process of DNN.

features. Following that, activation, feature abstraction, and non-linear transformations are fully connected hidden layers. Advanced Activation Functions for increased non-linearity, dropout layers to reduce overfitting, and batch normalization for stable learning to improve model performance are among the methods used in regularisation & optimization layers. The output layer generates a categorization label or anomaly score, providing interesting analysis. The DNNs-BCT framework's feature selection procedure aims to improve anomaly detection accuracy with little computational cost. The model leverages key characteristics collected from blockchain security events, device activity logs, and network traffic to zero in on the most important cyber danger indicators. Data exfiltration and distributed denial of service attacks can be detected using traffic-based features such as packet size distribution, flow duration, and protocol usage. Irregular communication patterns can be identified using statistical features such as network flows' mean, variance, and entropy. Behavioral features, such as failed authentication attempts, access pattern deviations, and anomalies in smart contract invocation, can be used to uncover Sybil and brute-force attacks, which are blockchain-based threats. To train DNNs with the most relevant network behaviors, the feature set is refined using methods like Recursive Feature Elimination (RFE), Principal Component Analysis (PCA), and Mutual Information-Based Selection. Most importantly, for applications such as anomaly detection, classification, and more, this ordered DL pipeline ensures efficient data processing, enabling models to grasp complex patterns and produce accurate predictions.

$$e_{de} [\partial' + us''] = b [s - fr''] + Na [w - uy''] \quad (9)$$

A threat management system that is both blockchain-secured  $b [s - fr'']$  and improved with AI processes  $Na [w - uy'']$  the discovered IoT anomalies, as shown by Eq. 9 ( $e_{de} [\partial' + us'']$ ). This proves that the suggested approach can secure IoT settings in realtime and is impermeable to cybersecurity.

$$p_{dr} [se - yw''] = [w - u''] + csx'' * Na [w - iu''] \quad (10)$$

A safe and adaptive threat response system is represented by Eq. 10, which involves evaluating risk behaviors in IoT networks ( $p_{dr} [se - yw'']$ ). A combination of blockchain technology ( $[w - u'']$ ) and artificial intelligence  $csx'' * Na$  improves the ability to detect  $[w - iu'']$ . This exemplifies the suggested method's capacity to offer robust, decentralized, and self-learning IoT protection against ever-changing cyber threats.

$$Bs = Vs [w - ua''] + B [a - sn''] * K [s - yur''] \quad (11)$$

A verified, AI-enhanced  $Vs [w - ua'']$ , decentralized threat administration system  $B [a - sn'']$  is created by transforming  $K [s - yur'']$  blockchain-secured IoT security procedures ( $Bs$ ) into the given Eq. 11. This exemplifies how the suggested approach may safeguard IoT networks with a solution that is resilient, flexible, and impossible to manipulate.

$$l_{se} [a - fj''] = N [s' + ur''] * B [u - sw''] + Jd [s - oi''] \quad (12)$$

Equation 12 shows  $l_{se} [a - fj'']$  how the security event logs of the (IoT) may be transformed into an anomaly detection system  $Jd [s - oi'']$  that is both blockchain-secured  $N [s' + ur'']$  and powered by artificial intelligence  $B [u - sw'']$ . The capability of the suggested technology to offer a self-learning, decentralized, and strong cybersecurity structure for IoT networks.

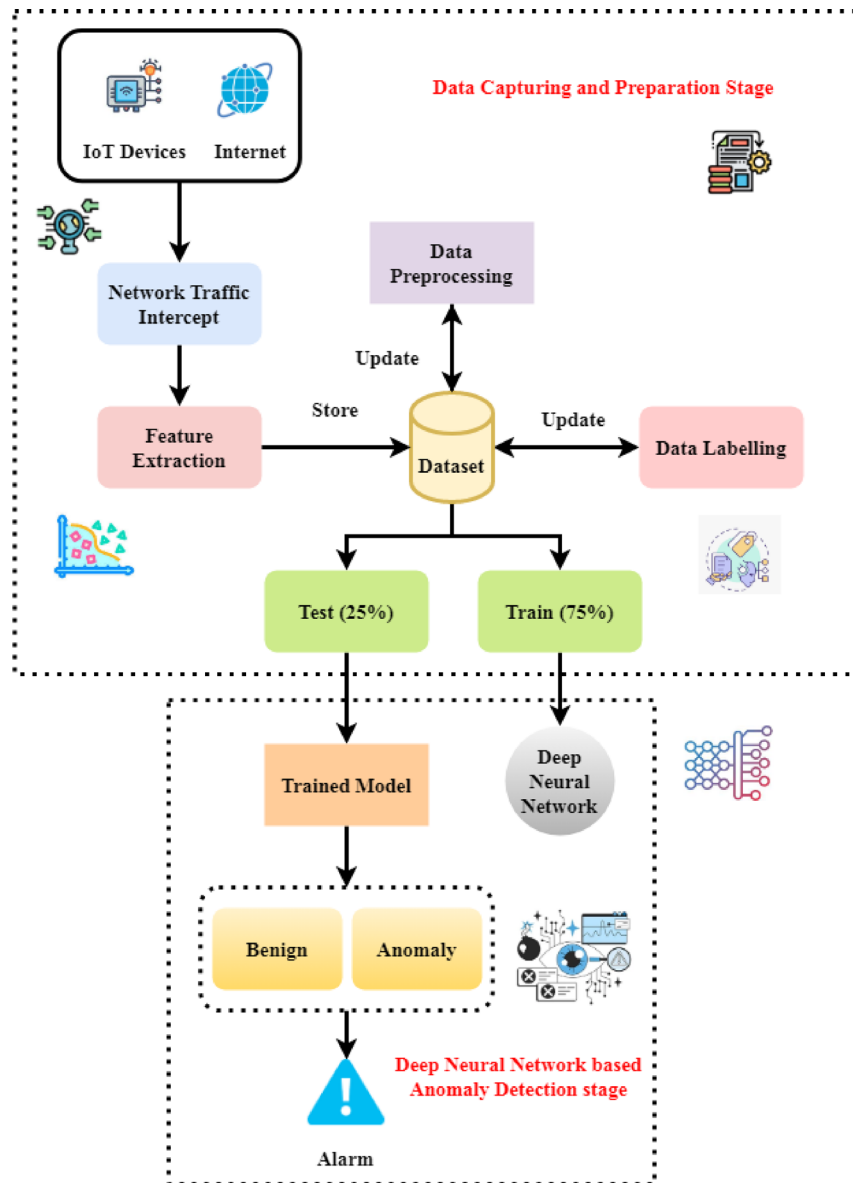
Figure 4 shows an IoT network traffic anomaly detection system based on DNN. It consists mostly of two main processes: data collecting and preprocessing and anomaly detection. The first stage intercepts network traffic and generates features. Data forms a dataset employing preprocessing and labeling. The next division of the dataset is 25% for testing and 75% for training. During the second step, a DNN learns to spot anomalies in the dataset. The trained model sets network behavior as either normal, benign, or aberrant, either benign or dangerous. An alert indicates a likely security risk upon an anomaly discovered. This technology guarantees IoT network integrity by employing machine learning techniques to spot suspicious activity in real-time, improving IoT security and helping to lower cyber risks.

$$c_zs = u [s - yr''] * js [e - jaw''] + Na [w - syu''] \quad (13)$$

Equation 13 shows how cybersecurity danger signals  $c_zs$  are transformed into an anomaly detection framework  $u [s - yr'']$  that is both blockchain-secured  $js [e - jaw'']$  and augmented  $Na [w - syu'']$  with artificial intelligence. This proves that the suggested approach can defend (IoT) networks with resilience, decentralization, and self-learning.

$$x_{vr} [w - au''] = Nx [s - dy''] + X [a - yr''] \quad (14)$$

The translation of IoT network vulnerabilities into a blockchain-secured as well as AI-driven threat detection system is represented by Eq. 14,  $x_{vr} [w - au'']$ . The AI-powered anomaly detection system  $Nx [s - dy'']$  is constantly learning and adjusting to new dangers, while the public ledger mechanism ( $X [a - yr'']$ ) guarantees secure, immutable authentication. Because of this, it is clear that the suggested approach may give an intelligent, decentralized, and strong cybersecurity framework to IoT networks.



**Fig. 4.** Training and testing flow in IoT network traffic anomaly detection system.

$$u_v s [i - an''] = Ba [w - ud''] + Ba [w - siu''] \quad (15)$$

Equation 15 shows  $u_v s [i - an'']$  how undiscovered security flaws in systems may be turned into an authentication framework protected by blockchain technology. Here, several layers  $Ba [w - siu'']$  and  $Ba [w - siu'']$ , which also prohibits unauthorized manipulation. This emphasizes the capacity of the suggested solution to improve IoT security using AI-driven risk identification and trust mechanisms based on blockchain.

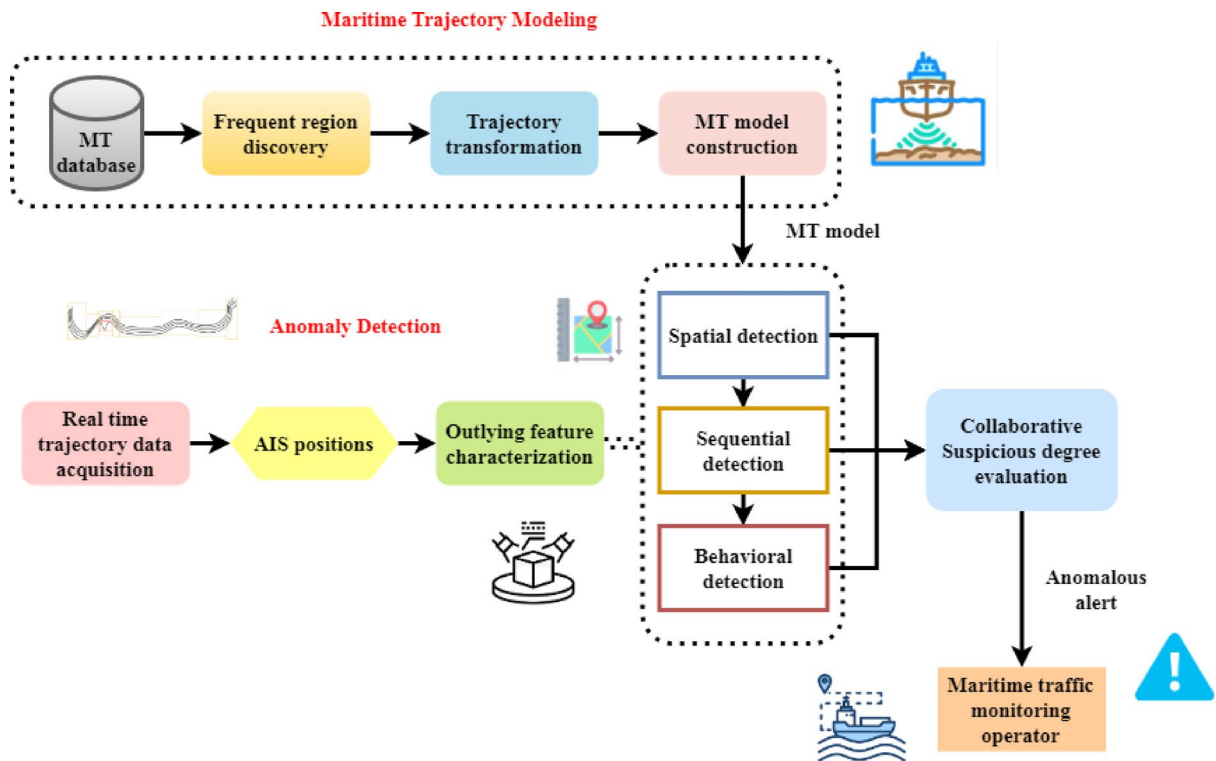
$$c_v w [i - sn''] = Ba [wl - sn''] + Ka [w - uy''] \quad (16)$$

To transform vulnerabilities in cybersecurity in IoT networks ( $c_v w [i - sn'']$ ) into a security framework supported by blockchain technology  $Ba [wl - sn'']$  and adaptive defensive mechanisms  $Ka [w - uy'']$  powered by artificial intelligence, Eq. 16 must be satisfied. The blockchain component guarantees data integrity and immutable logging, while AI-based anomaly detection constantly detects and mitigates risks.

### Contribution 3: Automated threat mitigation via smart contracts

Blockchain-based smart contracts for autonomous, real-time threat response provide safe, tamper-proof, distributed IoT security.

Figure 5 presents a grounded-on trajectory modeling maritime anomaly detecting system. It consists of two main phases: maritime trajectory modeling and anomaly detection. The first stage develops a Maritime



**Fig. 5.** Exploring marine trajectory behavior for anomalies.

Trajectory (MT) database employing the discovery of regular regions and changing trajectories. From this data, an MT model is developed, offering a foundation for anomaly identification in marine traffic. The second stage is obtaining real-time trajectory data using AIS (Automatic Identification System) sites. The method defines outlying properties using spatial, sequential, and behavioral detection to identify aberrant vascular movements. These results are even more investigated in the Collaborative Suspicious Degree Evaluation to assess the likelihood of unusual activity. Should an anomaly prove present, the operator of marine traffic monitoring gets a signal. Employing the identification of suspicious vessel behavior, this technology enhances marine security and helps authorities disrupt illegal operations, including smuggling and navigation.

$$p_{de} [k - sj''] = Ja [w - iu''] + Va [w - syu''] \quad (17)$$

The Eq. 17 shows  $p_{de}$  how possible data exposure networks may be transformed into a blockchain-protected authentication layer  $Ja [w - iu'']$  and anomaly detection  $Va [w - syu'']$  driven by artificial intelligence. As a result, the suggested method's capacity to improve IoT security via adaptive threat detection and decentralized data protection is demonstrated.

$$p_{sw} [i - st''] = Ls [w - sy''] * Ba [w - su''] \quad (18)$$

Equation 18 depicts the move from possible security holes in IoT networks ( $p_{sw} [i - st'']$ ) to a multi-level security system based on blockchain technology. The blockchain component guarantees decentralized access control  $Ba [w - su'']$ , while the AI model  $Ls [w - sy'']$  constantly scans network traffic. With this integration, (IoT) systems can better withstand ever-changing cyber threats.

$$p_{sde} = bvs [w - uy''] + a [s - ye''] * Bs [w - uy''] \quad (19)$$

A strong defense mechanism integrating behavioral verification systems ( $bvs [w - uy'']$ ), adaptive security analytics ( $a [s - ye'']$ ), and blockchain security ( $Bs [w - uy'']$ ) can be formed by transforming potential security shortages in IoT environments ( $p_{sde}$ ) into the given Eq. 19. This comprehensive method strengthens IoT networks by detecting cyber threats in real-time and enforcing security decentralization.

$$p_{de} = Va [w - sy''] + Hsp [si - sn] * Va [so - yt''] \quad (20)$$

Equation 20 shows the process by which possible security flaws  $p_{de}$  in the IoT settings are handled by conducting vulnerability assessments ( $Va [so - yt'']$ ) and implementing hybrid security measures ( $Hsp [si - sn]$ ), which

are then strengthened by further validation ( $Va[w - sy'']$ ). The combination guarantees enhanced security policies, strong protection of IoT networks, and real-time threat mitigation.

This paper covers anomaly detection led by AI in IoT networks, blockchain transactions, and marine security. Blockchain guarantees openness and confidence; DL models extract characteristics, categorize abnormalities, and trigger automatic reactions. These methods enhance danger identification and prevention, from monitoring vessel movements to protecting IoT communications. Integrating blockchain and AI strengthens cybersecurity, lowering risks and maintaining system integrity.

Misclassification of threats and decreased anomaly detection accuracy may occur due to adversarial assaults, in which small changes in input data trick DL networks. Implementing adversarial training, gradient masking, and detection techniques might strengthen the resistance of the DNN model against such assaults. Furthermore, weaknesses in smart contracts, Sybil attacks, and 51% assaults may still penetrate blockchain-based security procedures and jeopardize the network's consensus and integrity. Formal smart contract verification, reputation-based consensus methods, and lightweight cryptographic algorithms to guarantee blockchain security without excessive computational complexity should be included in the framework to avoid these dangers.

## Result and discussion

Among the key cybersecurity issues brought about by the fast spread of IoT devices are unauthorized access, malware assaults, and data breaches. Changing attack strategies and real-time threats test conventional security solutions. This paper suggests blockchain-based architecture and integrated DNN to ensure data integrity, enhance anomaly detection, and offer IoT networks autonomous, distributed threat-mitigating properties.

### Dataset description

The global digital security control system market is expected to grow from USD 15 billion in 2023 to USD 34 billion by 2032, with a CAGR of 9.5%. Three key drivers behind growth are rising cyberthreats, increasing IoT usage, and tight legal regulations. Among the technical breakthroughs are artificial intelligence, blockchain, and open security possibilities for machine learning<sup>26</sup>. North America dominates the sector, even though quick digitalization and rising cyberattacks will most likely force Asia Pacific to have the highest growth. CICIDS2017 records traffic from actual assaults such as DDoS, brute force, and penetration, while NSL-KDD gives organized data about attacks but has limitations because of old patterns. Ideal for real-time anomaly detection, UNSW-NB15 and Bot-IoT provide current IoT-specific threats, including spoofing, injection, and reconnaissance assaults. TON-IoT is a great tool for protecting vital infrastructure, healthcare facilities, and smart cities because it combines network telemetry with data from the industrial IoT. These datasets are valuable, but they have a class imbalance that makes it hard to generalize models since normal traffic is so prevalent. This is why DNNs-BCT use stratified sampling, anomaly-based augmentation, and the Synthetic Minority Over-sampling Technique (SMOTE) to improve learning. The model's capacity to react to new cyber threats is further enhanced by adding real-world IoT traffic records from smart home and industrial IoT contexts.

To better understand the relative performance of the framework, current assessments may depend on restricted baseline comparisons. All three types of intrusion detection systems (IDS) traditional machine learning (e.g., Random Forest, SVM, and KNN), DL (e.g., CNN, LSTM, and Transformer-based models), and security frameworks that use blockchain technology should be evaluated against the suggested model for a more thorough evaluation. The system's practical relevance and generalizability might be further enhanced by evaluations utilizing real-world IDS datasets such as CICIDS2017, NSL-KDD, or UNSW-NB15.

The suggested DNN-Blockchain architecture surpasses traditional rule-based intrusion detection systems (Fig. 6) with an astounding threat detection accuracy of 99.18%. This excellent accuracy of the DNN is driven by its ability to continuously learn from IoT traffic patterns and detect even the smallest changes connected with cyber threats. Using adaptive learning and multi-layer feature extraction, the model effectively identifies dangerous activities, including malware injections and DDoS attacks, ensuring powerful and consistent anomaly detection in IoT systems.

$$t_{da} = [au' + us''] + Va[s - sn''] * Y[w - su''] \quad (21)$$

By integrating advanced unsupervised training ( $[au' + us'']$ ), assessment of vulnerability ( $Va[s - sn'']$ ), and reinforcing mechanisms ( $Y[w - su'']$ ), the equation depicts the analysis of the detection of threats accuracy ( $t_{da}$ ). The method improves IoT security frameworks, reduces the number of false positives, and helps identify threats more effectively.

The biggest disadvantage of conventional intrusion detection systems is their great false-positive rate, resulting in useless alerts and inefficiencies. A low rate of 15.42%, shown in Fig. 7, the proposed DNN model significantly reduces false positives. This development is attributed to the adaptive learning process, as it improves classification accuracy over time. Reducing false alarms allows the system to increase operational efficiency, thereby allowing security staff to focus on actual threats and reduce unnecessary disturbance of IoT networks.

$$Fp = Ba[w - su''] + [a + ur''] + Ya[w - suy''] \quad (22)$$

The analysis of falsely positive results ( $Fp$ ) is captured by Eq. 22 by combining a Bayesian method ( $Ba[w - su'']$ ), learning that is adaptive ( $[a + ur'']$ ), and a validation mechanism based on reinforcement learning ( $Ya[w - suy'']$ ). This method improves system efficiency by eliminating unwanted alarms and ensuring high detection reliability.

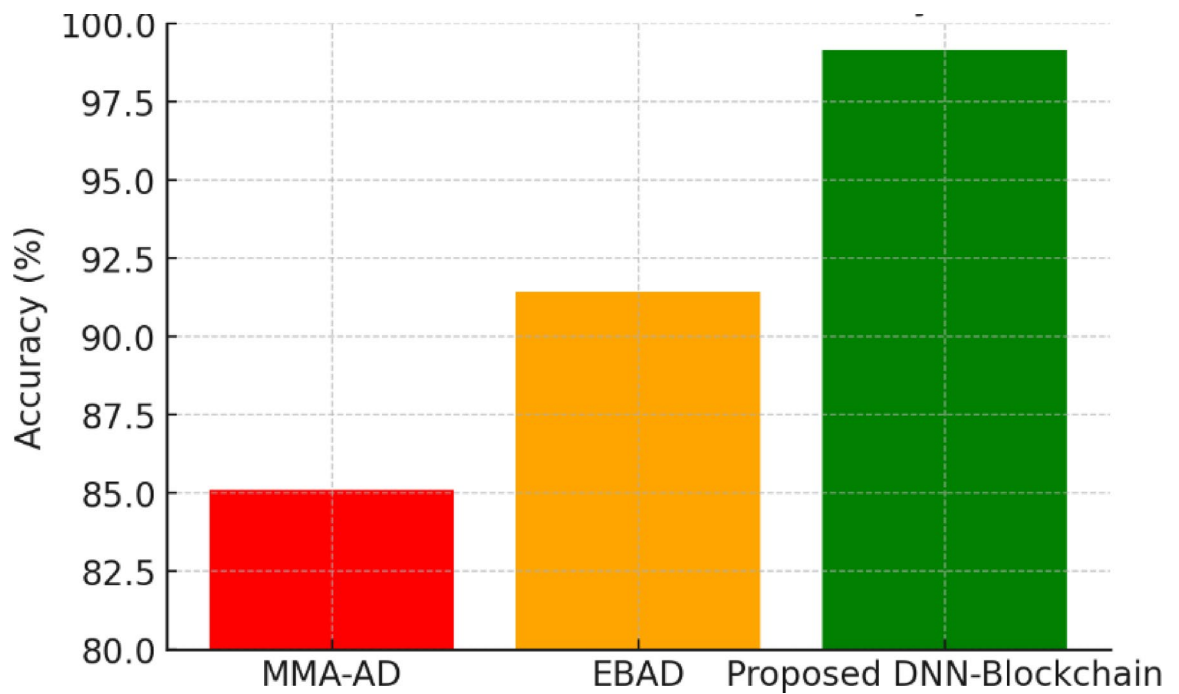


Fig. 6. Analysis of threat detection accuracy.

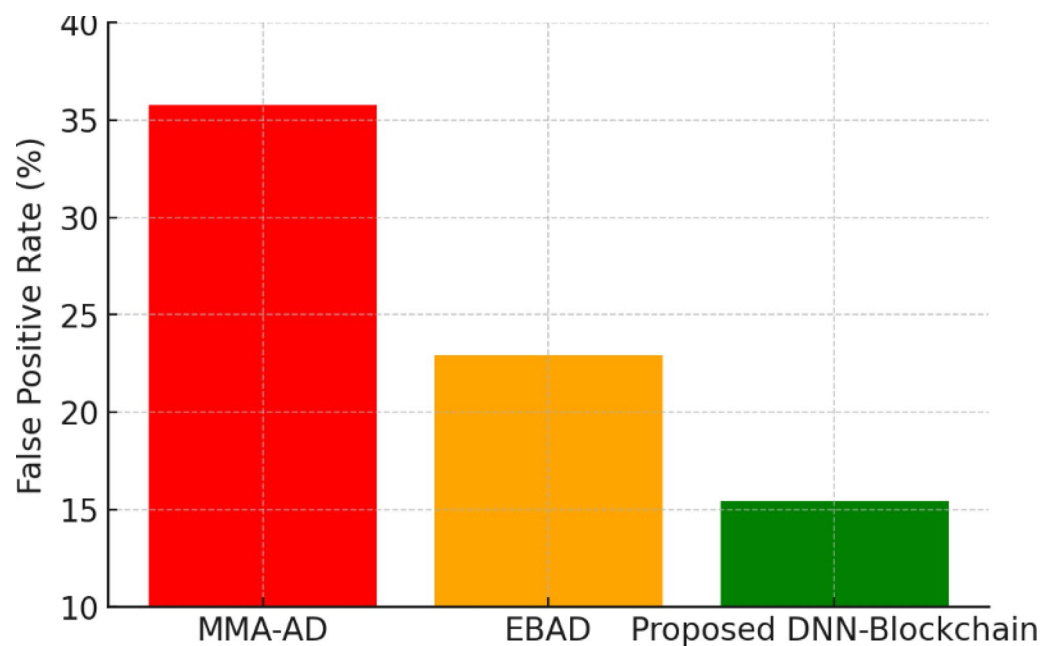


Fig. 7. Analysis of false-positive.

Showing a 95.25% efficiency in reaction time, the system ensures real-time threat mitigation (Fig. 8). Conventions in cybersecurity often suffer from delays in seeing and addressing anomalies, increasing the risk of damage. Smart contracts built within the Blockchain enable the system to react to threats automatically, therefore eliminating delays in human interaction. This fast decision-making ability enhances IoT security by rapidly neutralizing assaults, halting data breaches, and maintaining low latency linked device integrity preservation.

$$RT = [ui - ew''] + Ba [s - t''] + Ya [enk - ny''] \quad (23)$$

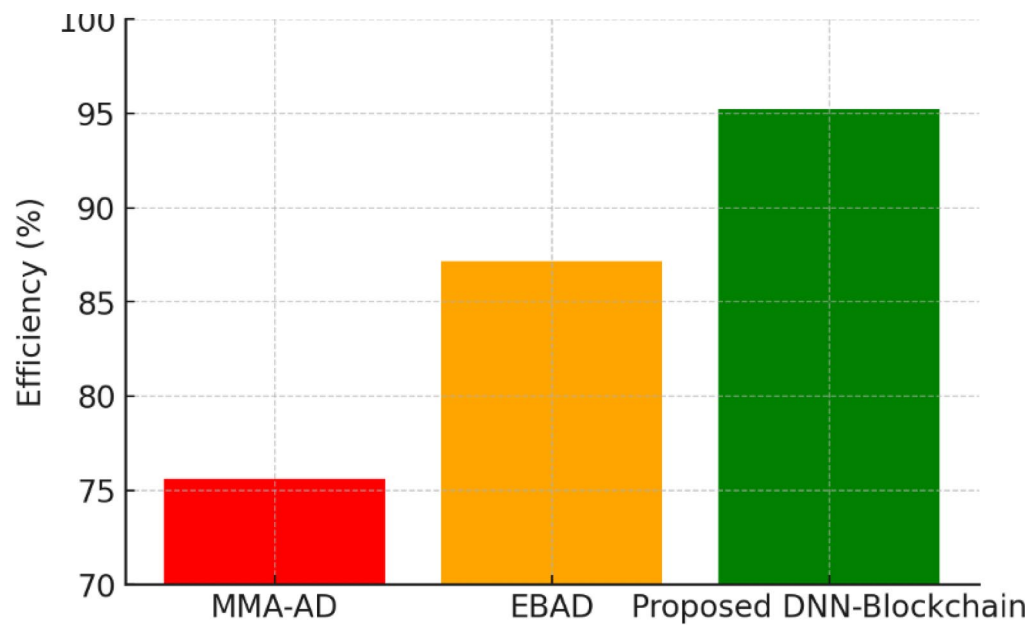


Fig. 8. Analysis of response time.

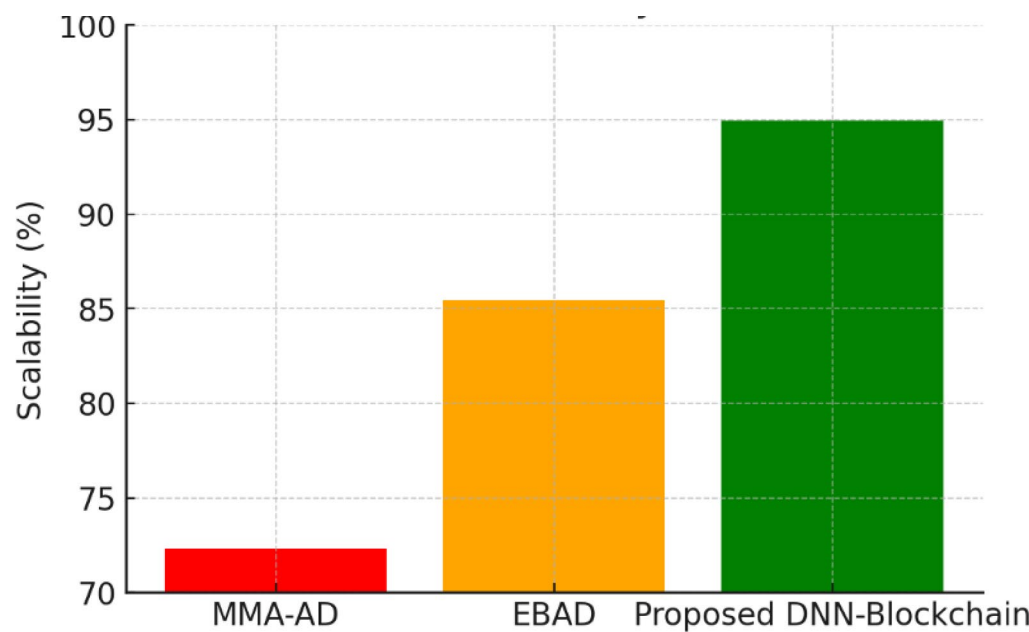


Fig. 9. Analysis of scalability.

Combining input from users' delay ( $[ui - ew'']$ ), Bayesian-based planning ( $Ba[s - t']$ ), and an AI-driven enrichment mechanism ( $Ya[enk - ny'']$ ) into the analysis of response time ( $RT$ ) is what Eq. 23 represents. This method keeps cybersecurity threat mitigation very accurate while minimizing system latency. Distributed storage alternatives like IPFS or cloud-based systems may be used off-chain to store less sensitive data, reducing transaction overhead. The blockchain should only record key security events, including identified abnormalities or authentication failures. Further ways to improve performance and decrease blockchain bloat include batching transactions and compressing logs. To minimize delays caused by latency, the system should optimize the execution of smart contracts so that automatic answers and real-time verification are possible. Edge computing may preprocess IoT data before blockchain integration to further reduce network congestion. Guaranteeing perfect interaction with the huge IoT systems shown in Fig. 9, the framework ensures a 94.96% scalability. Unlike centralized security solutions, which struggle with growing network size and data

loads, our distributed approach efficiently handles ever more linked devices. Blockchain's distributed ledger for secure transactions, paired with the DNN model's ability to accommodate massive IoT traffic volumes without performance degradation, ensures that the solution stays trustworthy and efficient even as the IoT network expands.

$$S = pd_{se} [d - qa''] + [yt - uj''] + Bsw [oi - sn''] \quad (24)$$

Incorporating parallel distributed system performance  $pd_{se} [d - qa'']$ , dynamic load adaptability  $[yt - uj'']$ , and blockchain-supported balanced workloads  $Bsw [oi - sn'']$ , the equation represents the evaluation of scalability ( $S$ ). The method improves the system's scalability by using decentralized computing methodologies and optimizing the allocation of resources in Eq. 24.

The method significantly lowers single points of failure by using a decentralization score of 97.33%, enhancing IoT network security (Fig. 10). While the Blockchain-integrated solution ensures distributed, tamper-proof data storage and authentication, conventional centralized security systems are open to deliberate access. Every IoT device logs activities on a private Blockchain, ensuring trust and preventing illicit alterations. This significant degree of decentralization guarantees its self-sustaining character and strengthens the IoT ecosystem by improving resilience against cyberattacks, openness, and security.

$$D = Bs [w - uy''] + Ua [w - sy''] - hfs'' \quad (25)$$

By combining blockchain-supported processes ( $Bs [w - uy'']$ ) and unified self-governing systems ( $Ua [w - sy'']$ ), the Eq. 25 stands for the analysis of decentralization ( $D$ ) and the reduction of centralized control factors ( $hfs''$ ). The method uses blockchain technology and autonomous decision-making systems to bolster decentralization, as shown in Table 3.

The hybrid DNN-Blockchain architecture for IoT network security is presented in this study. Blockchain guarantees automatic threat reaction and tamper-proof security; the DNN detects abnormalities with 99.18% accuracy. The system has tremendous scalability—94.96%, responds rapidly (95.25%), decentralizes (97.33%), and reduces false positives (15.42%). Results show how well IoT ecosystems are locked by artificial intelligence-blockchain synergy, as shown in Table 3 and Fig. 11.

A comprehensive computational cost analysis for DNN training and blockchain operations is necessary to guarantee that the DNNs-BCT system is feasible for real-world deployment, especially in IoT contexts with limited resources. To decrease inference delay and energy consumption, optimization methods like pruned architectures, quantization, or knowledge distillation may be necessary for DNN models, which are useful for anomaly detection but computationally costly. When assessing the training phase, it is important to consider the memory footprint, processing time, and GPU/TPU needs to guarantee scalability. The blockchain adds processing costs with transaction verification, block production, and consensus procedures. Although blockchain technology improves security by recording transactions in an immutable way and allowing for decentralized authentication, the limited resources of IoT devices require cryptographic techniques that are efficient and

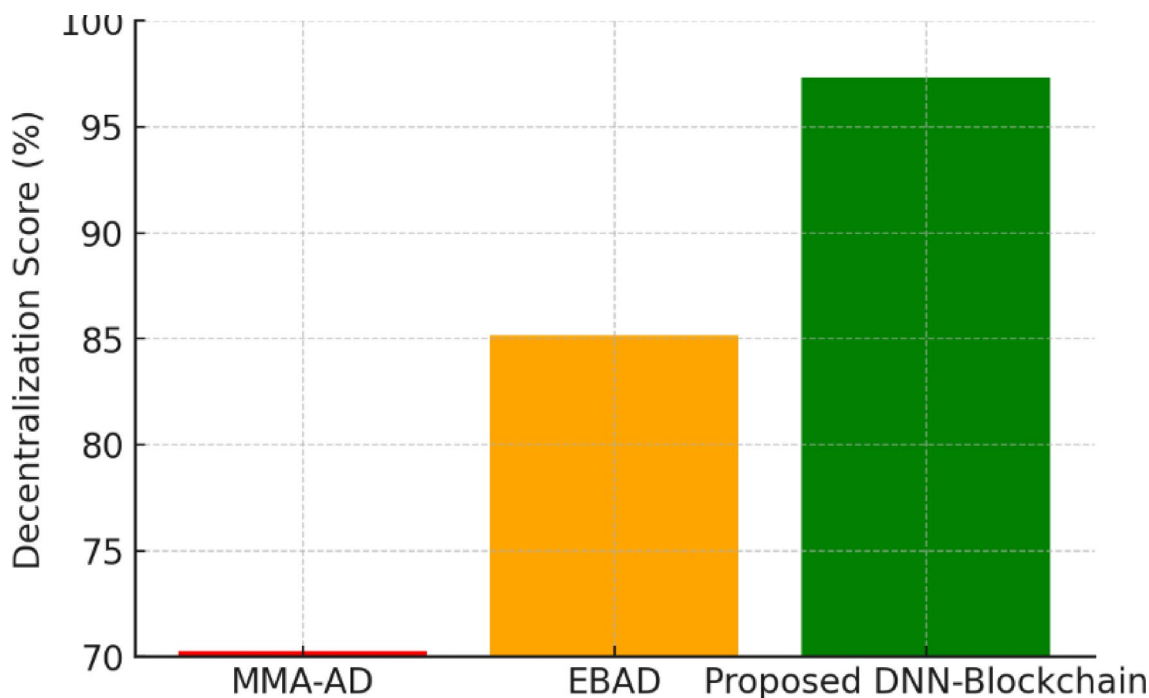
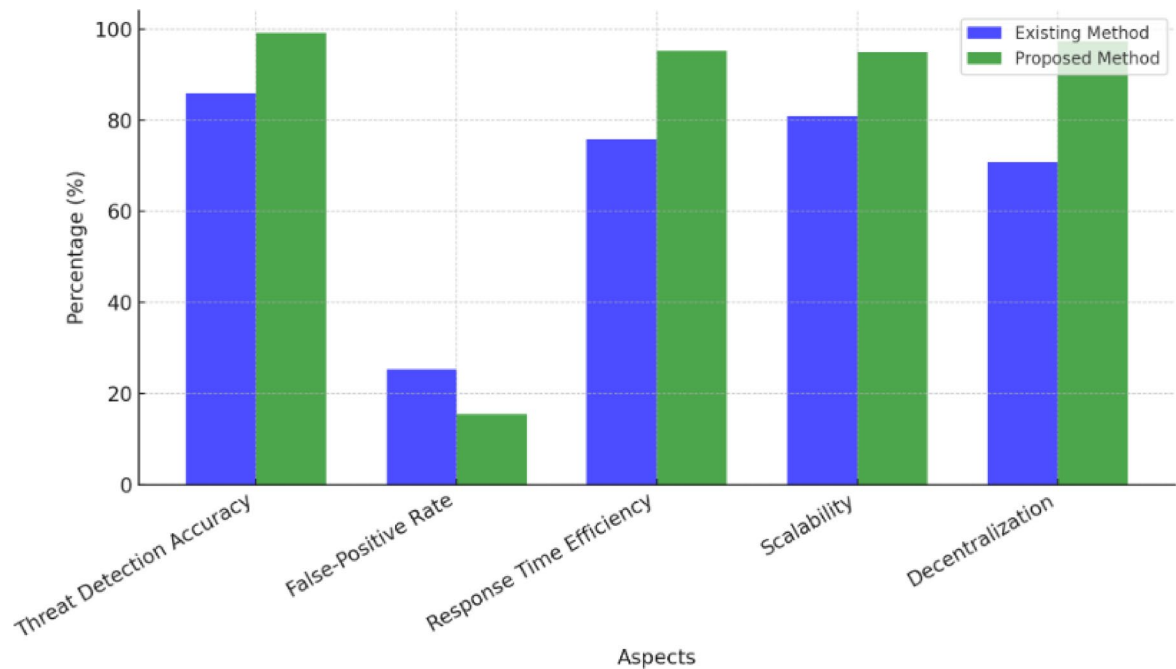


Fig. 10. Analysis of decentralization.

| Aspects                   | Existing method in ratio (%) | Proposed method in ratio (%) | Key features                                                                                        |
|---------------------------|------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------|
| Threat detection accuracy | 85.92                        | 99.18                        | High-precision DNN-based anomaly detection with adaptive learning                                   |
| False-positive rate       | 25.35                        | 15.42                        | Significant reduction in false alarms, improving reliability                                        |
| Response time efficiency  | 75.85                        | 95.25                        | Real-time threat mitigation using smart contracts on Blockchain                                     |
| Scalability               | 80.88                        | 94.96                        | Efficient handling of large-scale IoT networks with minimal performance degradation                 |
| Decentralization          | 70.83                        | 97.33                        | Blockchain integration ensures tamper-proof, distributed security without central points of failure |

**Table 3.** Comparison of existing method and proposed method.



**Fig. 11.** Comparison of existing and proposed method.

lightweight, such as elliptic curve cryptography or zero-knowledge proofs, to keep power consumption and storage needs to a minimum.

**Conclusion**

Rising security concerns in IoT systems demand a sophisticated security architecture adept at real-time anomaly identification and prevention. This paper offers a hybrid method combining DNNs with blockchain technologies to enhance IoT security. With a low false-positive rate of 15.42% and a strong detection accuracy of 99.18%, the proposed model successfully identifies malicious activity, including malware injections and Distributed Denial of Service (DDoS) assaults. Blockchain technology replaces single points of failure and forbids illegal changes by providing data integrity, openness, and decentralizing powers. Furthermore, smart contracts allow autonomous, real-time attack responses, enhancing reaction time efficiency (95.25%) and general system scalability (94.96%). The experimental findings confirm that the DNN-Blockchain synergy exceeds conventional security methods in accuracy, speed, and durability. The distributed character of blockchain enhances trust and authentication, so IoT systems become safer against growing cyberattacks. This study stresses the requirement of adaptive learning motivated by artificial intelligence and tamper-proof security systems in IoT settings. Built on the recommended framework, future cybersecurity solutions also provide the means for self-learning, autonomous, scalable IoT security architectures competent in real-time adaptation to new hazards. Future work will focus on lightweight DNN architectures perfect for IoT devices with constrained resources to raise efficiency without compromising security. Moreover, under investigation, federated learning will promote scalability and decentralization even because of data privacy preservation. Investigating including quantum-resistant cryptographic techniques into Blockchain will ensure long-term security against new hazards. These advances will strengthen IoT security, hence increasing network resilience and adaptability.

### **Data availability**

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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## Author contributions

Sathyabama A R: Problem Selection, Algorithm, Implementation, Coding and Testing Results. Jeevaa Katiravan.: Methodology, Formal Analysis, Supervision.

## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

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